

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 70%. We keep your highest score.

Next item →

1. Which of the following is correct?

1 / 1 point

- ☐ Evolutionary algorithms do not need a population.
 - ☒ For evolutionary algorithms, multiple runs may obtain different results.
 - ☐ Evolutionary algorithms guarantee obtaining the global optimum.
 - ☐ For evolutionary algorithms, multiple runs always obtain the same result.
- ☒ Nice work

2. Which of the following is correct?

1 / 1 point

- ☐ Binary string is the only candidate solution representation method in GA.
 - ☐ GA is a local search algorithm.
 - ☒ The crossover rate in GA should be large.
 - ☐ The mutation rate in GA should be large.
- ☒ Nice work

3. In genetic algorithms, the role of the crossover function is to _____.

1 / 1 point

- ☐ select Parent
 - ☐ mutate
 - ☐ evaluate fitness
 - ☒ generate new individuals
- ☒ Nice work

4. The main components of a genetic algorithm include _____.

1 / 1 point

- ☒ population initialization
- ☒ Nice work
- ☒ fitness evaluation
- ☒ Nice work
- ☒ crossover
- ☒ Nice work
- ☒ mutation
- ☒ Nice work
- ☒ selection
- ☒ Nice work

5. The basic idea of a genetic algorithm is to simulate _____.

1 / 1 point

- ☐ swarm intelligence
 - ☒ natural selection
 - ☐ gene mutation
- ☒ Nice work